



Australian
National
University

Weighted Randomness

Learn how to make random choices where some options are more likely than others — an operation at the core of all generative AI.

You will need

- 10-sided dice (d10)
- coloured marbles or beads in a bag

Your goal

To randomly choose from a fixed set of outcomes according to a given probability distribution.

Key idea

Sometimes we need to make random choices where some outcomes are more likely than others. There are ways to do this which ensure certain relationships *on average* between the outcomes (e.g. one outcome happening twice as often as another one).



Algorithm 1: beads in a bag

- **materials:** coloured beads, bag
- **setup:** count out a number of beads corresponding to the desired weights for each outcome
- **sampling procedure:** shake the bag, then draw one bead without looking

Example

You want to choose an ice cream flavour: **vanilla** 50% of the time, **chocolate** 30%, and **strawberry** 20%.

- add 5 white beads to the bag (corresponding to **vanilla**)
- add 3 brown beads to the bag (corresponding to **chocolate**)
- add 2 red beads to the bag (corresponding to **strawberry**)

Draw a bead from the bag — that's your ice-cream choice for today.

Algorithm 2: dice with ranges

- **materials:** d10 (or d6, d20 as alternatives)
- **setup:** assign number ranges proportional to weights (see table, right)
- **sampling procedure:** roll the die, then look up the corresponding outcome

Example

- for 60% vanilla/40% chocolate, roll a d10: 1-6 means **vanilla**, 7-10 means **chocolate**
- for 50% vanilla/30% chocolate/20% strawberry, roll a d10: 1-5 means **vanilla**, 6-8 means **chocolate**, 9-10 means **strawberry**

You can use different dice (d6, d10, d20, d120, etc.), it will just change the number ranges corresponding to each outcome.

d10 dice roll → outcome mapping table

	1				2					
2	0			4	5			9		
	1			2		3				
3	0		3	4		6	7		9	
	1		2		3		4			
4	0		2	3		5	6	7	8	9
	1		2		3		4		5	
5	0	1	2	3	4	5	6	7	8	9
	1	2	3	4	5	6	7	8	9	10
6	0	1	2	3	4	5	6	7	8	9
	1	2	3	4	5	6	7	8	9	10
7	0	1	2	3	4	5	6	7	8	9
	1	2	3	4	5	6	7	8	9	10
8	0	1	2	3	4	5	6	7	8	9
	1	2	3	4	5	6	7	8	9	10
9	0	1	2	3	4	5	6	7	8	9